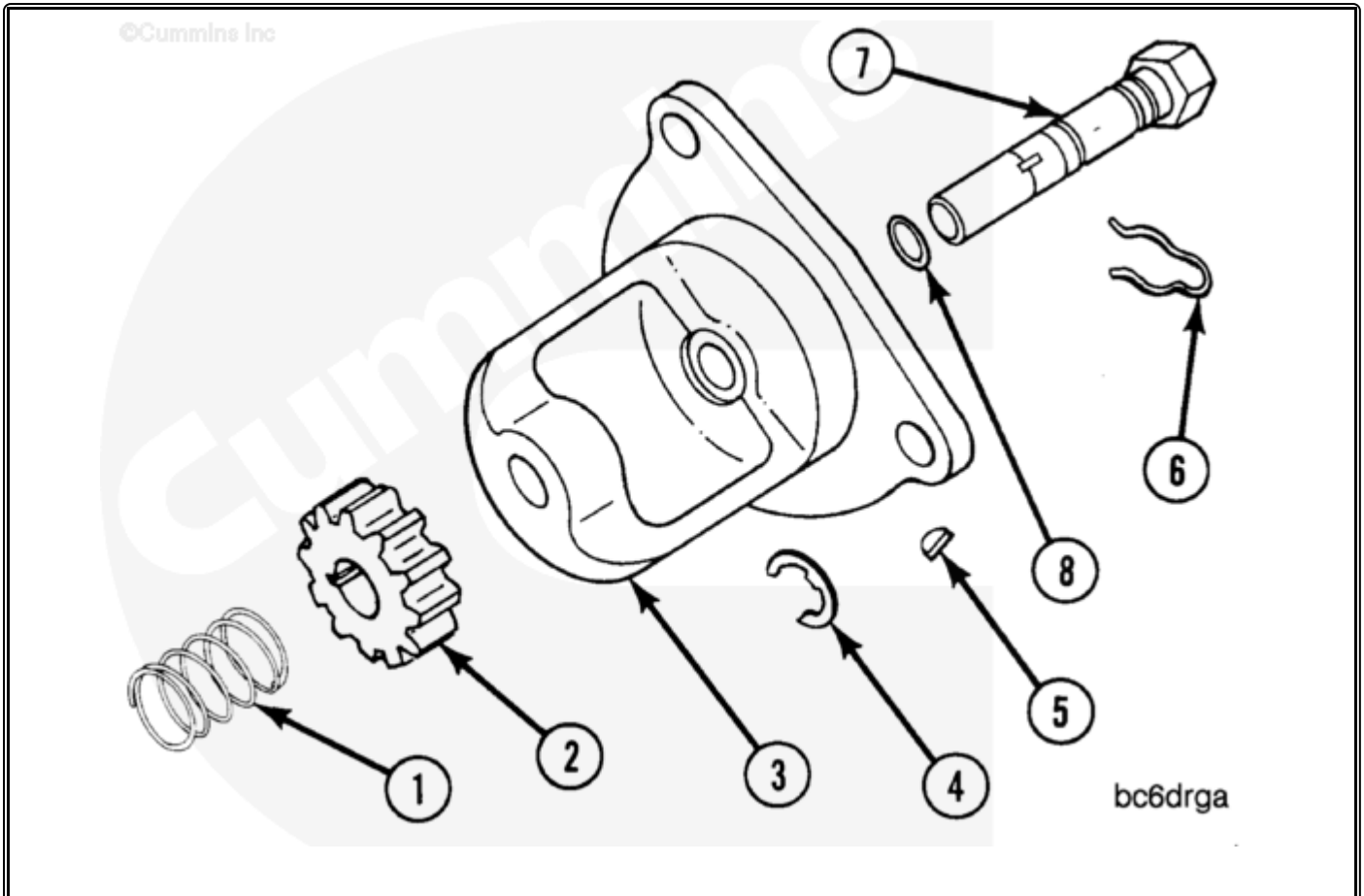


Trainings ▾

Exploded View



1. Compression spring
2. Engine barring device gear
3. Engine barring device housing
4. Retaining E-ring
5. Woodruff key
6. External cotter clip
7. Engine barring device shaft
8. O-ring.

Remove

Note : Only engines equipped with wet type flywheel housings have the gaskets (5).

Remove the starter cover capscrews (7) and lock washers (8).

Remove the starter cover (6) and gasket (5).

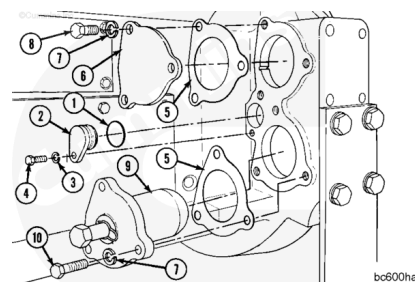
Remove the access hole cover capscrew (4) and lock washer (3).

Remove the access hole cover (2), and remove the o-ring (1) from the access cover.

Remove the engine barring device capscrews (10) and lock washers (7).

Remove the engine barring device (9), and gasket (5).

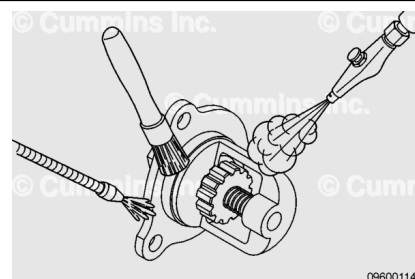
Discard the gaskets and o-ring.



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

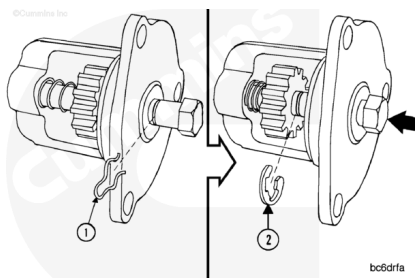
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the barring device with solvent and dry with compressed air.

Disassemble

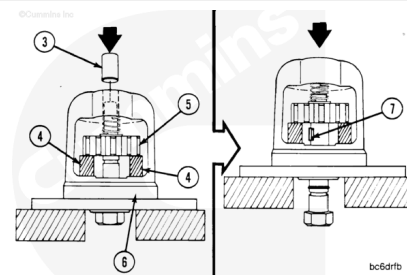
Remove the external cotter clip (1).

Push the shaft in and remove the retaining E-ring (2).



⚠ CAUTION ⚠

Do not attempt to press the shaft out with spacers or supports between the gear and the housing. The key in the shaft will damage the housing.



Place the engine barring device in a arbor press.

The supports **must** be a minimum of 21 mm [0.813 in] long to allow the key to clear the gear and **not** contact the housing.

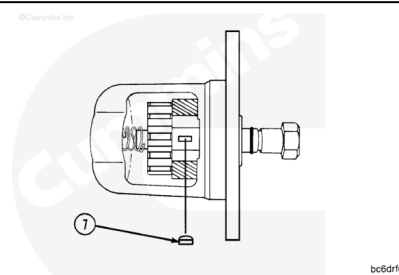
A steel pipe coupling, size 3.18 mm [0.125 in], has been used successfully for the supports.

Place supports (4) between the gear (5) and the housing (6).

Push the shaft in until the key (7) is exposed.

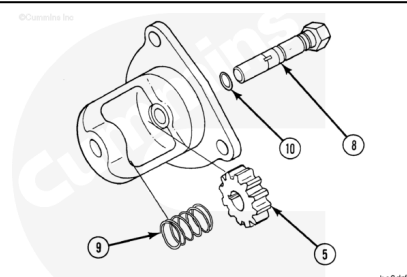
Remove the key (7).

Remove the supports.



Remove the following:

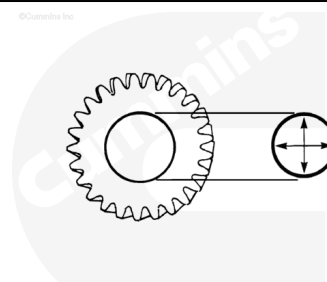
- Shaft (8)
- Spring (9)
- Gear (5)
- O-ring (10).



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the parts with solvent and dry with compressed air.

Check the gear teeth for damage and wear.

Replace the gear if the teeth are damaged.

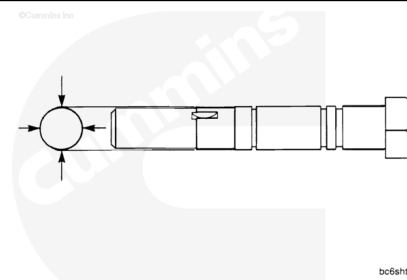
Measure the inside diameter of the gear.

Engine Barring Device Gear Inside Diameter		
mm		in
18.77	MIN	0.739
18.80	MAX	0.740

If the gear is **not** within specifications, the gear **must** be replaced.

Measure the outside diameter gear area of the shaft.

Engine Barring Device Shaft Outside Diameter		
mm		in
18.82	MIN	0.741
18.84	MAX	0.742



If the shaft is **not** within specifications, the shaft **must** be replaced.

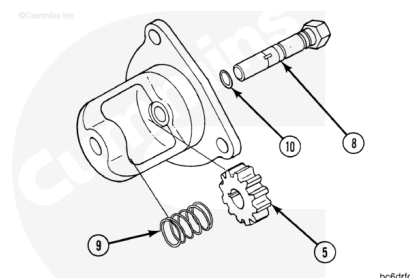
Assemble

Install the o-ring (10) into the second groove from the hexagon head end of the shaft (8).

Lubricate the o-ring and shaft with clean engine oil.

Install the gear (5) and spring (9) into the housing. The spring will secure the gear in the housing.

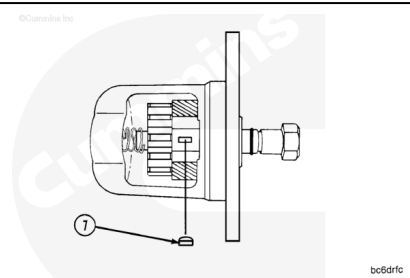
Install the shaft (8) through the housing and in the gear until the shoulder touches the gear.



Push in the shaft until the keyway is exposed.

Install the key (7) into the shaft.

Align the keyway in the gear with the key.



⚠ CAUTION ⚠

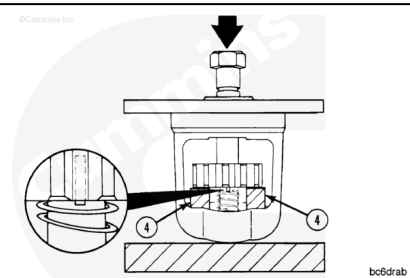
Make sure the key is aligned with the keyway in the gear before pressing the shaft into the gear. If the key is not aligned properly, damage to the engine barring device will result.

Place the engine barring device in an arbor press with the hexagon side of the shaft positioned up.

Install the supports (4) into the housing as shown in the illustration.

Press the shaft into the gear until the end of the key is even with the face of the gear.

Remove the supports.

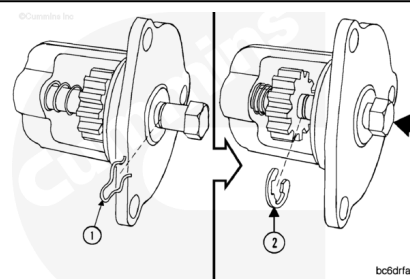


Push the shaft in until the groove for the E-ring is exposed.

Install the E-ring (2).

Allow the shaft to extend until the E-ring rests against the housing.

Install the external cotter clip (1).



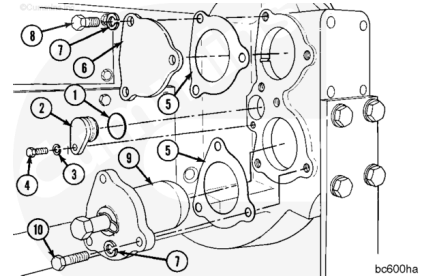
Install

Install the o-ring seal (1) onto the access hole cover (2).

Lubricate the o-ring seal with vegetable oil.

Install the access hole cover (2) in the bore in the flywheel housing.

Install the lock washer (3) and capscrew (4) and tighten the capscrew.



Torque Value: 9 n•m [80 in-lb]

Note : Gaskets are not required on a dry type flywheel housing.

Install the gasket (5), cover plate (6), lock washers (7), and capscrews (8).

Tighten the capscrews.

Torque Value: 60 n•m [44 ft-lb]

Install the gasket (5), engine barring device (9), lock washers (7), and capscrews (10).

Tighten the capscrews.

Torque Value: 185 n•m [136 ft-lb]